Wireless Local Area Networks
Mạng cục bộ không dây

Tuấn

October 27, 2026

Abstract

This document covers the definition and evolution of Wi-Fi, the spectrum and channel plans, network architecture and deployment models, and security. Appendices provide a list of abbreviations and an English–Vietnamese terminology table.

Contents

1 Introduction **3**
- 1.1 What Is a WLAN? . . . 3
- 1.2 LAN and WLAN . . . 3
- 1.3 WLAN Compared with Other Wireless Networks . . . 3
- 1.4 Advantages and Limitations . . . 3

2 IEEE 802.11 Standards and Spectrum **4**
- 2.1 Generations of Wi-Fi . . . 4
- 2.2 Functional Amendments . . . 4
- 2.3 Frequency Bands and Channels . . . 6
  - 2.3.1 The 2.4 GHz band . . . 6
  - 2.3.2 The 5 GHz band . . . 6
  - 2.3.3 The 6 GHz band . . . 6
  - 2.3.4 Other bands . . . 7
- 2.4 Regulatory . . . 7

3 WLAN Architecture and Components **7**
- 3.1 Basic Building Blocks . . . 7
- 3.2 Service Sets and Topologies . . . 8
- 3.3 Deployment Models . . . 9
- 3.4 Network Services Around the WLAN . . . 9

4 WLAN Security **10**
4.1 Why Wireless Needs Strong Security . . . 10
4.2 Generations of Wi-Fi Security . . . 10
4.3 WPA2 and the 4-Way Handshake . . . 10
4.4 WPA3 . . . 11
4.5 Enterprise Authentication . . . 12
4.6 Threats and Countermeasures . . . 12
4.7 Hardening . . . 12

A List of Abbreviations **14**

B English–Vietnamese Terminology **16**

List of Figures

1 2.4 GHz channel plan: channels 1, 6 and 11 (bold) do not overlap. 6
2 An Extended Service Set: two BSSs joined by a wired distribution system. . . . 8
3 The WPA2/WPA3 4-way handshake. . . . 11

List of Tables

1 Classes of wireless networks (typical values). . . . 3
2 Wi-Fi generations and PHY amendments. . . . 5
3 802.11 functional amendments. . . . 5
4 Common WLAN deployment models. . . . 9
5 Generations of WLAN security. . . . 10
6 Common EAP methods. . . . 12
7 Common WLAN threats and defences. . . . 13
8 Core English–Vietnamese WLAN terminology. . . . 16

1 Introduction

1.1 What Is a WLAN?

A Wireless Local Area Network (WLAN) is a data network that connects devices over a radio-frequency medium instead of cables, typically inside a home, an office floor, a campus, or a public venue. Modern WLANs are almost exclusively built on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards, which define the Physical (layer) (PHY) and Medium Access Control (MAC) layers [IEE21].

1.2 LAN and WLAN

A LAN connects devices within a limited area, usually using switched Ethernet and physical cables. A WLAN provides the same local-network function over a shared wireless medium, giving users mobility and easier deployment but making performance more sensitive to distance, interference and obstacles. A WLAN normally connects to the rest of the LAN through an access point.

1.3 WLAN Compared with Other Wireless Networks

Wireless networks are commonly classified by coverage area (Table 1). WLANs sit between short-range personal networks and wide-area cellular networks.

Table 1: Classes of wireless networks (typical values).

Class	Typical range	Examples	Spectrum	Typical use
Wireless Personal Area Network (WPAN)	up to about 10 m	Bluetooth	Unlicensed (2.4 GHz)	Headsets, sensors, wearables
WLAN	tens of metres indoors	Wi-Fi (802.11)	Unlicensed 2.4, 5, 6 GHz	Homes, offices, campuses
Wireless MAN	several kilometres	WiMAX (802.16)	Licensed and unlicensed	Fixed broadband, backhaul
Wireless Wide Area Network (WWAN)	kilometres per cell	LTE	Licensed	Mobile broadband, voice

1.4 Advantages and Limitations

In normal use, a station associates with an access point and exchanges frames over the shared wireless medium. This model is used in homes, offices, campuses, public hotspots and temporary networks where mobility or avoiding new cabling is important.

- **Advantages:**
  - **Mobility and convenience:** users keep connectivity while moving between rooms or buildings.
  - **Fast, low-cost deployment:** no cable pulling; ideal for temporary sites, heritage buildings and dense venues.
  - **Scalability:** capacity grows by adding Access Points (APs) and channels.
- **Disadvantages:**
  - **Shared medium:** all stations in range of the same channel compete for the same airtime, so capacity is lower and less predictable than with switched Ethernet.
  - **Interference and propagation effects:** walls, reflections and other radios degrade performance.
  - **Security exposure:** radio signals leave the building, so strong encryption and authentication are mandatory.

Tóm tắt

- WLAN là mạng cục bộ dùng sóng vô tuyến thay cho cáp, hầu hết xây dựng trên họ chuẩn IEEE 802.11 (định nghĩa lớp PHY và MAC).
- So với LAN có dây, WLAN linh hoạt và dễ triển khai hơn nhưng dùng chung môi trường truyền, nên nhạy với khoảng cách, nhiễu và vật cản.
- Về phạm vi, WLAN nằm giữa WPAN (khoảng 10 m) và WWAN (hàng km mỗi ô).
- Di động, dễ triển khai nhanh và rẻ, mở rộng bằng cách thêm AP và kênh. Nhưng có ít dung lượng để chia sẻ, dễ nhiễu, sóng lan ra diện rộng nên bắt buộc phải có mã hóa và xác thực mạnh.

---

2 IEEE 802.11 Standards and Spectrum

2.1 Generations of Wi-Fi

Table 2 summarises the main PHY amendments. Generation names for 802.11b/a/g are informal and retroactive.

2.2 Functional Amendments

Besides PHY amendments, many amendments add MAC-level features. The most relevant ones are listed in Table 3.

Table 2: Wi-Fi generations and PHY amendments.

Generation / amendment	Year	Bands (GHz)	Max width (MHz)	Max QAM	Peak PHY rate	Key technologies
– / 802.11	1997	2.4	22	–	2 Mb/s	FHSS, DSSS, infrared; CSMA/CA
– / 802.11b	1999	2.4	22	–	11 Mb/s	HR-DSSS with CCK
– / 802.11a	1999	5	20	64	54 Mb/s	OFDM (52 subcarriers)
– / 802.11g	2003	2.4	20	64	54 Mb/s	OFDM in 2.4 GHz, compatible with 11b
Wi-Fi 4 / 802.11n	2009	2.4, 5	40	64	600 Mb/s	MIMO (4 streams), channel bonding, A-MPDU
Wi-Fi 5 / 802.11ac	2013	5	160	256	6.93 Gb/s	8 streams, downlink MU-MIMO, beamforming
Wi-Fi 6/6E / 802.11ax	2021	2.4, 5, 6	160	1024	9.6 Gb/s	OFDMA, UL/DL MU-MIMO, BSS Colouring, TWT
Wi-Fi 7 / 802.11be	2024	2.4, 5, 6	320	4096	about 46 Gb/s	MLO, multi-RU, preamble puncturing

Table 3: 802.11 functional amendments.

Amendment	Topic	Purpose
802.11e	QoS	Prioritised channel access (EDCA), basis of WMM
802.11h	Spectrum management	DFS and TPC for coexistence with radar in 5 GHz
802.11i	Security	CCMP/AES, 4-way handshake (WPA2)
802.11k	Radio resource measurement	Neighbour reports and beacon measurements to assist roaming
802.11r	Fast BSS Transition	Reduces handoff delay by pre-deriving keys
802.11v	BTM	Network can suggest a better AP to a client
802.11w	PMF	Authenticates deauthentication/disassociation frames
802.11s	Mesh networking	Multi-hop forwarding between mesh APs
802.11u	Interworking	Basis of Passpoint/Hotspot 2.0 roaming agreements
802.11ai	FILS	Fast initial authentication and link setup
802.11ad / ay	60 GHz (DMG / EDMG)	Short-range multi-gigabit links
802.11ah	Sub-1 GHz (Wi-Fi HaLow)	Long-range, low-power Internet of Things (IoT)
802.11az	Positioning	Improved ranging and location accuracy
802.11ba	Wake-up radio	Ultra-low-power wake-up for Internet of Things (IoT) devices
802.11bf	WLAN sensing	Using Wi-Fi signals for presence and motion detection

2.3 Frequency Bands and Channels

2.3.1 The 2.4 GHz band

Spanning roughly 2400–2483.5 MHz, the 2.4 GHz Industrial, Scientific and Medical (ISM) band offers good range and wall penetration but is narrow and crowded (Bluetooth, Zigbee, microwave ovens, cordless phones). Channels are 5 MHz apart but a 20/22 MHz signal overlaps about four adjacent channel numbers. Channels 1, 6 and 11 are the usual non-overlapping planning choice. In Vietnam, the 2.4 GHz band utilizes channels 1–13 and equipment must comply with the Ministry of Information and Communications (MIC) technical standards, such as QCVN 54:2020/BTTTT, to determine permitted maximum transmission power.

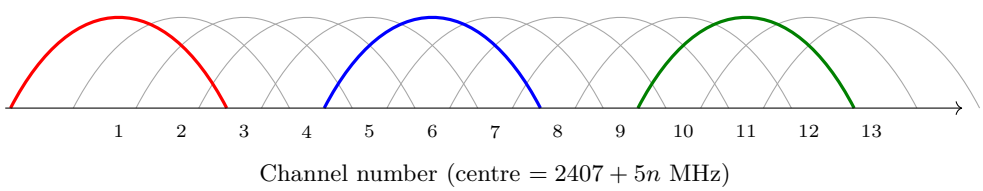


Figure 1: 2.4 GHz channel plan: channels 1, 6 and 11 (bold) do not overlap.

2.3.2 The 5 GHz band

The 5 GHz band is divided into sub-bands with different indoor, outdoor and power conditions. Some channels require Dynamic Frequency Selection (DFS): an AP must listen for radar and vacate the channel if radar is detected. In Vietnam, 5 GHz WLAN deployments are strictly regulated by the MIC; hardware must meet specific QCVN standards for conformity certification before clearing customs or deployment. The centre frequency of a 5 GHz channel n is

$$f_c = 5000 + 5n \text{ MHz}. \tag{1}$$

Channels may be bonded into 40, 80 and 160 MHz widths, but wide channels reduce the number of independent channels and increase the risk of co-channel interference.

2.3.3 The 6 GHz band

Wi-Fi 6E and Wi-Fi 7 can use the 6 GHz band. Under Vietnam's Circular 01/2025/TT-BKHCN (effective 15 May 2025), the 5925–6425 MHz spectrum is officially allocated for Wi-Fi. Indoor usage is allowed up to 200 mW Equivalent Isotropically Radiated Power (EIRP) (max Power Spectral Density (PSD) of 10 dBm/MHz), while outdoor deployments are limited to 25 mW EIRP (max PSD of 1 dBm/MHz). Devices are forbidden on Unmanned Aircraft Systemss

(UASs). Where 6 GHz operation is permitted, WPA3 (or Opportunistic Wireless Encryption (OWE)) is mandatory.

2.3.4 Other bands

The 60 GHz band (802.11ad/ay) offers very high rates over a few metres in a single room. Sub-1 GHz (802.11ah, "HaLow") trades speed for range and wall penetration for Internet of Things (IoT) sensors.

2.4 Regulatory

In Vietnam, the MIC sets national frequency allocation and technical conformity (QCVN) standards determining permitted WLAN channels, transmit power, indoor and outdoor use, and DFS requirements. Devices must undergo mandatory conformity testing and obtain certification by submitting a Declaration of Conformity (DoC).

Tóm tắt

- Wi-Fi phát triển từ 802.11 (1997, 2 Mb/s) đến Wi-Fi 7 (802.11be: kênh 320 MHz, 4096-QAM, MLO, tốc độ PHY lý thuyết tối đa khoảng 46 Gb/s). Tốc độ đỉnh chỉ là lý thuyết.
- Các bản bổ sung chức năng cho PHY: QoS (e), DFS/TPC (h), bảo mật (i, w), chuyển vùng (k, r, v), mesh (s), cùng các bản cho IoT, định vị và cảm biến.
- Băng 2,4 GHz: tầm xa, xuyên tường tốt nhưng hẹp và đông đúc; kênh 1, 6, 11 không chồng lấn; Việt Nam dùng kênh 1–13.
- Băng 5 GHz: nhiều kênh hơn, có thể ghép 40/80/160 MHz, một số kênh yêu cầu DFS để tránh radar.
- Băng 6 GHz: theo Thông tư 01/2025/TT-BKHCN, dải 5925–6425 MHz dành cho Wi-Fi; trong nhà tối đa 200 mW EIRP, ngoài trời 25 mW, không được sử dụng trên UAS; bắt buộc WPA3 hoặc OWE.
- Thiết bị tại Việt Nam phải tuân thủ các quy chuẩn QCVN do MIC quy định, được kiểm tra hợp quy và nộp bản DoC.

---

3 WLAN Architecture and Components

3.1 Basic Building Blocks

Station
: Any device with an 802.11 radio: laptop, phone, IoT sensor or printer. A client STA associates with an AP.

Access Point
: Bridges wireless clients to the wired LAN. An AP may host several virtual networks, each with its own Service Set Identifier (SSID).

Wireless LAN Controller
Centralises configuration, Radio Resource Management (RRM), security policy and roaming for many lightweight APs.

Distribution System
The (usually wired Ethernet) backbone that interconnects APs and provides access to servers and the Internet.

Power over Ethernet
Standards 802.3af (15 W), 802.3at (30 W), and 802.3bt (60–90 W) deliver power to APs over the Ethernet cable; modern Wi-Fi 6/7 APs may need 802.3at or 802.3bt.

3.2 Service Sets and Topologies

- **Basic Service Set (BSS):** one AP and its associated stations. Identified by the Basic Service Set Identifier (BSSID), normally the AP radio's MAC address.
- **Independent Basic Service Set (IBSS):** an ad hoc peer-to-peer network without an AP.
- **Extended Service Set (ESS):** several BSSs connected by a DS and sharing an SSID, allowing roaming.
- **Mesh BSS:** APs forward traffic wirelessly to each other (802.11s or vendor mesh); only some nodes need a wired uplink.

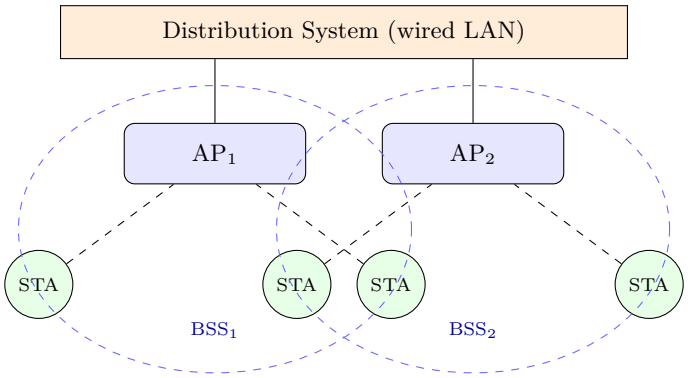


Figure 2: An Extended Service Set: two BSSs joined by a wired distribution system.

Table 4: Common WLAN deployment models.

Model	Control plane	Data plane	Strengths	Weaknesses
Autonomous (standalone) APs	Each AP	Local switching	Simple, cheap for 1–3 APs	Hard to scale and manage consistently
Controller-based (central)	WLC	Tunnelled to WLC via CAPWAP	Central policy, easy L3 roaming	Controller can be a bottleneck
Controller-based (local switching)	WLC	Local at AP (e.g. FlexConnect-style)	Good for branches, survives WAN loss	More complex VLAN design
Cloud-managed	Cloud service	Local at AP	Zero-touch provisioning, analytics	Management depends on Internet
Mesh	Distributed or controller	Multi-hop wireless	No cabling to every AP	Throughput drops per hop

3.3 Deployment Models

In controller-based designs, Control And Provisioning of Wireless Access Points (CAPWAP) [CMS09b, CMS09a] carries control messages (UDP port 5246, protected by DTLS) and optionally client data (UDP port 5247) between lightweight APs and the WLC.

3.4 Network Services Around the WLAN

- **VLANs and SSIDs:** each SSID is normally mapped to a Virtual Local Area Network (VLAN) (corporate, guest, voice, IoT) so traffic can be segmented and policed.
- **DHCP and DNS:** must be sized for the peak number of clients and short lease times in guest networks.
- **RADIUS:** authentication, authorisation and accounting for 802.1X (Section 4).
- **Layer-2 vs Layer-3 roaming:** within one subnet a client keeps its IP address; across subnets a controller either tunnels traffic to an anchor or the client must obtain a new IP address (via DHCP).

Tóm tắt

- Thành phần cơ bản: STA (trạm), AP (điểm truy cập), WLC (bộ điều khiển), DS (hệ thống phân phối) và PoE (cấp nguồn qua cáp Ethernet).
- Các tập dịch vụ: BSS (một AP và các trạm của nó), IBSS (ad hoc, không AP), ESS (nhiều BSS nối qua DS, cho phép chuyển vùng) và Mesh BSS.
- Mô hình triển khai: AP độc lập, dùng bộ điều khiển (chuyển mạch tập trung hoặc cục bộ), quản lý qua đám mây và mesh; mỗi mô hình cân bằng giữa tính đơn giản, khả năng mở rộng và mức độ phụ thuộc.
- CAPWAP (UDP 5246 cho điều khiển, 5247 cho dữ liệu) kết nối AP nhẹ với WLC.
- Dịch vụ xung quanh: ánh xạ SSID–VLAN để phân đoạn lưu lượng, DHCP/DNS, RADIUS cho 802.1X, và chuyển vùng lớp 2/lớp 3.

4 WLAN Security

4.1 Why Wireless Needs Strong Security

Anyone within radio range can capture frames. A secure WLAN must therefore provide *confidentiality* (encryption), *integrity* (tamper detection), *authentication* (who is connecting, and to what), and *availability* (resistance to jamming and management-frame attacks).

4.2 Generations of Wi-Fi Security

Table 5: Generations of WLAN security.

Mechanism	Year	Cipher / key management	Status
WEP	1997	RC4, 24-bit IV, static key	Broken; keys recoverable in minutes [FMS01]. Never use.
WPA	2003	RC4 with TKIP, PSK or 802.1X	Deprecated interim fix.
WPA2	2004	AES-CCMP, PSK or 802.1X	Widely deployed; offline guessing of weak passphrases and KRACK-type flaws.
WPA3	2018	AES-CCMP/GCMP, SAE or 802.1X, mandatory PMF	Current recommendation [Wi-18].

4.3 WPA2 and the 4-Way Handshake

In WPA2-Personal the passphrase (8–63 characters) and SSID are fed into PBKDF2 to produce the 256-bit Pairwise Master Key (PMK):

$$\mathrm{PMK} = \mathrm{PBKDF2}\big(\text{HMAC-SHA1},\ \text{passphrase},\ \text{SSID},\ 4096,\ 256\big). \tag{2}$$

In WPA2/WPA3-Enterprise the PMK comes from 802.1X/Extensible Authentication Protocol (EAP) authentication instead. The 4-way handshake (Figure 3) proves both sides know the PMK and derives fresh session keys, the Pairwise Transient Key (PTK), from the PMK, both MAC addresses and two random numbers (nonces):

$$\mathrm{PTK} = \mathrm{PRF}\big(\mathrm{PMK},\ \text{“Pairwise key expansion”},\ \mathrm{AA} \,\|\, \mathrm{SPA} \,\|\, \mathrm{ANonce} \,\|\, \mathrm{SNonce}\big), \tag{3}$$

where AA and SPA are the authenticator and supplicant MAC addresses, taken in a fixed min/max order so both sides compute the same value. The PTK is split into a key-confirmation key, a key-encryption key and a temporal key used by Counter Mode with CBC-MAC Protocol (CCMP). Message 3 also delivers the Group Temporal Key (GTK) for broadcast/multicast traffic, encrypted and protected by a Message Integrity Code (MIC).

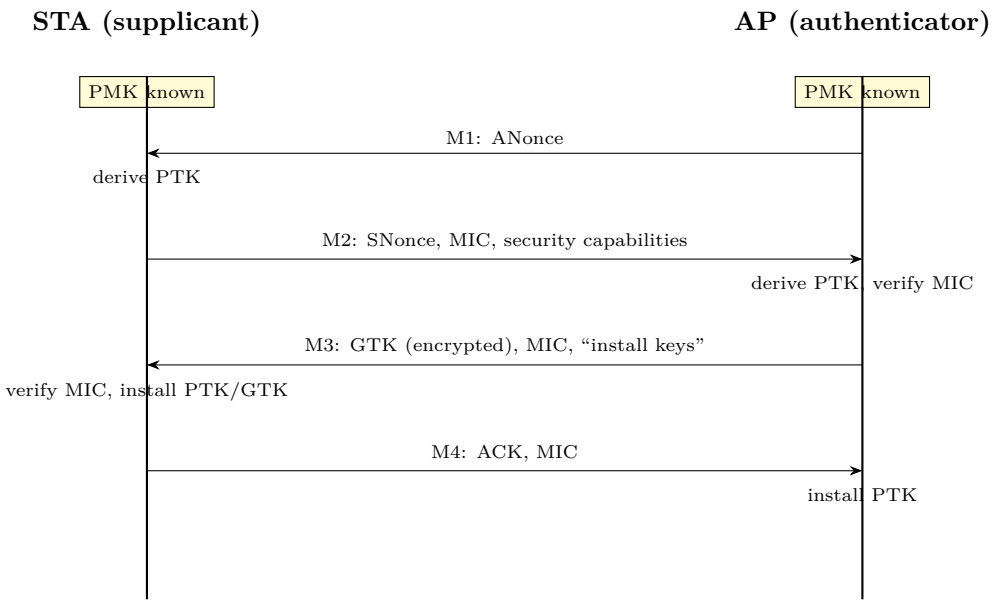


Figure 3: The WPA2/WPA3 4-way handshake.

4.4 WPA3

- **Simultaneous Authentication of Equals (SAE)** replaces the Pre-Shared Key (PSK) exchange with a password-authenticated key exchange. Captured handshakes cannot be attacked offline and each guess requires interaction with the AP; keys also gain forward secrecy.
- **Protected Management Frames (PMF)** (802.11w) is mandatory, protecting deauthentication and disassociation frames against forgery.

- **WPA3-Enterprise 192-bit mode** uses Galois/Counter Mode Protocol (GCMP)-256 and 384-bit elliptic-curve cryptography for high-security environments.
- **OWE ("Enhanced Open")** [HK17] encrypts open networks (e.g. cafés) without a password via an unauthenticated key exchange, defeating passive eavesdropping.
- **Device Provisioning Protocol (DPP)** (Wi-Fi Easy Connect) onboards headless IoT devices with QR codes or similar.
- Early WPA3 implementations had side-channel and downgrade weaknesses ("Dragonblood") that were addressed by updates [VR20]; keep firmware current.

The 6 GHz band requires WPA3 (or OWE), so Wi-Fi 6E/7 networks are secure by default.

4.5 Enterprise Authentication

IEEE 802.1X [IEE20] defines three roles: the *supplicant* (client), the *authenticator* (AP or switch) and the *authentication server*, typically a RADIUS server [RWRS00]. The authenticator blocks all traffic except EAP over LAN (EAPOL) until the server returns success; the EAP [ABV+04] method carries the credentials inside a Transport Layer Security (TLS) tunnel where applicable.

Table 6: Common EAP methods.

Method	Credentials	Notes
EAP-TLS	Client and server certificates	Strongest; requires a PKI and certificate distribution
PEAP / EAP-TTLS	Server certificate; user name and password inside tunnel	Common; must validate the server certificate to avoid credential theft
EAP-FAST	Protected access credential	Cisco-originated; avoids certificates
EAP-SIM / AKA	SIM card	Mobile-operator Wi-Fi offload

4.6 Threats and Countermeasures

4.7 Hardening

- Use WPA3 (or WPA2-AES only); disable WEP, WPA/TKIP, and WPS.
- Prefer 802.1X with certificates for corporate networks; use passphrases of at least 16 randomly chosen characters for personal/PSK networks.
- Enable PMF; keep AP/controller firmware up to date.

Table 7: Common WLAN threats and defences.

Threat	Description	Countermeasure
Eavesdropping	Passive capture of frames on open or weakly encrypted networks	WPA2/WPA3, OWE for open guest networks
Rogue AP	Unauthorised AP attached to the wired network	WIPS, wired port security, regular scans
Evil twin	Fake AP imitating a legitimate SSID to trap clients	802.1X with server-certificate validation, WIPS
Deauthentication / disassociation flood	Forged management frames force clients offline	PMF (802.11w), WPA3
Offline passphrase guessing	A captured WPA2 handshake tested against dictionaries	Long random passphrases, SAE, 802.1X
Key reinstallation (KRACK)	Replaying handshake messages to reset nonces [VP17]	Patched clients and APs
WPS PIN abuse	Short PIN of Wi-Fi Protected Setup brute-forced	Disable WPS
Weak segmentation	Compromised guest or IoT device reaches internal systems	VLANs, firewall policy, client isolation

- Separate corporate, guest, voice and IoT traffic into different VLANs with firewall rules; enable client isolation on guest SSIDs.
- Deploy WIPS or built-in rogue detection; monitor logs for repeated authentication failures.
- Do not rely on hidden SSIDs or MAC filtering; they offer negligible security and are trivially bypassed.

Tóm tắt

- Ai trong vùng phủ sóng cũng có thể bắt khung, nên WLAN cần đảm bảo bảo mật, toàn vẹn, xác thực và tính sẵn sàng.
- Tiến hóa: WEP (đã bị phá), WPA/TKIP, WPA2 (AES-CCMP, bắt tay bốn bước; còn rủi ro dò mật khẩu ngoại tuyến và KRACK), WPA3.
- WPA3: SAE thay PSK để chống dò ngoại tuyến, PMF bắt buộc, chế độ Enterprise 192-bit, OWE mã hóa mạng mở, DPP đăng ký thiết bị IoT.
- Doanh nghiệp dùng 802.1X (supplicant, authenticator, máy chủ RADIUS) với EAP; EAP-TLS mạnh nhất, PEAP/EAP-TTLS phổ biến nhưng phải kiểm tra chứng chỉ máy chủ.
- Các mối đe dọa chính: nghe lén, rogue AP, evil twin, tấn công deauthentication, dò mật khẩu, KRACK, lạm dụng WPS, phân đoạn yếu.
- Củng cố: dùng WPA3 (hoặc WPA2-AES), tắt WEP/TKIP/WPS, bật PMF, dùng mật khẩu ngẫu nhiên dài từ 16 ký tự, phân tách VLAN, triển khai WIPS và cập nhật firmware. SSID ẩn và lọc MAC không phải là biện pháp bảo mật.

A List of Abbreviations

AES Advanced Encryption Standard

AP Access Point

BSS Basic Service Set

BSSID Basic Service Set Identifier

CAPWAP Control And Provisioning of Wireless Access Points

CCMP Counter Mode with CBC-MAC Protocol

DFS Dynamic Frequency Selection

DoC Declaration of Conformity

DPP Device Provisioning Protocol

DS Distribution System

EAP Extensible Authentication Protocol

EAPOL EAP over LAN

EIRP Equivalent Isotropically Radiated Power

ESS Extended Service Set

GCMP Galois/Counter Mode Protocol

GTK Group Temporal Key

IBSS Independent Basic Service Set

IEEE Institute of Electrical and Electronics Engineers

IoT Internet of Things

ISM Industrial, Scientific and Medical

KRACK Key Reinstallation Attack

LAN Local Area Network

MAC Medium Access Control

MIC Ministry of Information and Communications

MIC Message Integrity Code

MLO Multi-Link Operation

OWE Opportunistic Wireless Encryption

PEAP Protected EAP

PHY Physical (layer)

PMF Protected Management Frames

PMK Pairwise Master Key

PoE Power over Ethernet

PSD Power Spectral Density

PSK Pre-Shared Key

PTK Pairwise Transient Key

QCVN Quy chuẩn kỹ thuật quốc gia (National Technical Regulation)

QoS Quality of Service

RADIUS Remote Authentication Dial-In User Service

RRM Radio Resource Management

SAE Simultaneous Authentication of Equals

SSID Service Set Identifier

STA Station

TKIP Temporal Key Integrity Protocol

TLS Transport Layer Security

TPC Transmit Power Control

UAS Unmanned Aircraft Systems

VLAN Virtual Local Area Network

WEP Wired Equivalent Privacy

WIPS Wireless Intrusion Prevention System

WLAN Wireless Local Area Network

WLC Wireless LAN Controller

WPA Wi-Fi Protected Access

WPAN Wireless Personal Area Network

WPS Wi-Fi Protected Setup

WWAN Wireless Wide Area Network

B English–Vietnamese Terminology

Table 8: Core English–Vietnamese WLAN terminology.

English term (abbr.)	Tiếng Việt	Short definition
Networks and architecture		
Local Area Network (LAN)	Mạng cục bộ	Network serving a limited geographic area
Wireless Local Area Network (WLAN)	Mạng cục bộ không dây	Local network using radio instead of cables
Wi-Fi	Mạng không dây Wi-Fi	Wi-Fi Alliance brand for certified 802.11 products
Access Point (AP)	Điểm truy cập	Device that bridges wireless clients to a wired network
Wireless station (STA)	Trạm không dây	Device with an 802.11 radio
Wireless LAN Controller (WLC)	Bộ điều khiển mạng không dây	Central manager for lightweight APs
Basic Service Set (BSS)	Tập dịch vụ cơ bản	One AP and its associated stations
BSS Identifier (BSSID)	Mã định danh BSS	Identifier, normally a MAC address, for a BSS
Extended Service Set (ESS)	Tập dịch vụ mở rộng	Several BSSs sharing one SSID
Service Set Identifier (SSID)	Tên mạng	Name advertised by a wireless network
Association	Liên kết	Procedure by which a station joins an AP
Distribution System (DS)	Hệ thống phân phối	Backbone interconnecting APs
Roaming	Chuyển vùng	Moving between APs without losing connectivity
Power over Ethernet (PoE)	Cấp nguồn qua Ethernet	Power delivered over the data cable
Virtual LAN (VLAN)	Mạng LAN ảo	Logical segmentation of a network
Radio and physical layer		
Wireless medium	Môi trường truyền không dây	Shared radio path used to exchange frames
Frequency	Tần số	Number of wave cycles per second
Band	Băng tần	Range of frequencies used by a radio system
Channel	Kênh	Portion of a band used by one network
Channel width	Độ rộng kênh	Spectrum occupied by a channel

continued on next page

Table 8 (continued)

English term (abbr.)	Tiếng Việt	Short definition
ISM band	Băng tần công nghiệp, khoa học và y tế	Unlicensed band such as 2.4 GHz
Regulatory domain	Miền quản lý tần số	Country-specific channel and power rules
Antenna gain	Độ lợi anten	Focusing ability of an antenna relative to isotropic radiation
Transmit power	Công suất phát	Radio power output, commonly measured in dBm
EIRP	Công suất bức xạ đẳng hướng tương đương	Transmit power plus antenna gain minus cable loss
RSSI	Cường độ tín hiệu thu	Measure of received signal level
Signal-to-Noise Ratio (SNR)	Tỷ số tín hiệu trên nhiễu	Signal level relative to background noise
Interference	Nhiễu	Unwanted signal degrading reception
Path loss	Suy hao đường truyền	Loss of power over distance
Multipath	Đa đường	Signal arriving over several propagation paths
Fading	Pha-đinh	Rapid signal variation caused by propagation effects
Modulation	Điều chế	Encoding data onto a carrier wave
Coding rate	Tốc độ mã hóa	Fraction of transmitted bits carrying information
MCS	Sơ đồ điều chế và mã hóa	Index selecting modulation and coding rate
OFDM	Ghép kênh phân chia theo tần số trực giao	Many orthogonal subcarriers in one channel
OFDMA	Đa truy cập phân chia theo tần số trực giao	Subcarriers allocated to different users
MIMO	Đa đầu vào đa đầu ra	Multiple antennas at transmitter and receiver
Beamforming	Tạo búp sóng	Focusing signal energy towards a client
Throughput	Thông lượng	Actual data delivered per second
Latency	Độ trễ	Time for data to travel from source to destination

MAC and quality of service

Medium Access Control (MAC)	Điều khiển truy cập môi trường truyền	Layer arbitrating use of the shared channel

continued on next page

Table 8 (continued)

English term (abbr.)	Tiếng Việt	Short definition
CSMA/CA	Đa truy cập tránh xung đột	Listen-before-talk access with random backoff
Carrier sense	Cảm nhận sóng mang	Detecting whether the medium is busy
Backoff	Thời gian lùi	Random wait before transmitting
Contention Window (CW)	Cửa sổ tranh chấp	Range from which backoff is chosen
Request to Send / Clear to Send (RTS/CTS)	Yêu cầu gửi / cho phép gửi	Handshake reserving the channel
Network Allocation Vector (NAV)	Vectơ phân bổ mạng	Timer for virtual carrier sensing
Frame	Khung	Basic MAC data unit transmitted over the air
Beacon	Khung báo hiệu	Periodic broadcast announcing a BSS
Authentication	Xác thực	Proving identity to the network
Acknowledgement (ACK)	Xác nhận	Frame confirming successful reception
Frame aggregation	Gộp khung	Combining several payloads in one transmission
Quality of Service (QoS)	Chất lượng dịch vụ	Prioritising traffic classes
Access Category (AC)	Loại truy cập	Voice, video, best-effort or background class
Power save mode	Chế độ tiết kiệm năng lượng	Radio sleeps between transmissions
Security		
Confidentiality	Tính bí mật	Data readable only by authorised parties
Integrity	Tính toàn vẹn	Data not altered undetected
Encryption	Mã hóa	Transforming data to an unreadable form
Wired Equivalent Privacy (WEP)	Bảo mật tương đương mạng có dây	Obsolete, broken wireless protection
Wi-Fi Protected Access (WPA)	Truy cập Wi-Fi được bảo vệ	Family of Wi-Fi security certifications
Advanced Encryption Standard (AES)	Chuẩn mã hóa nâng cao	Standard block cipher used by WPA2/WPA3
Pre-Shared Key (PSK)	Khóa chia sẻ trước	Passphrase-based authentication

continued on next page

Table 8 (continued)

English term (abbr.)	Tiếng Việt	Short definition
Simultaneous Authentication of Equals (SAE)	Xác thực đồng đẳng	WPA3 password-based key exchange
Protected Management Frames (PMF)	Khung quản lý được bảo vệ	Authenticated management frames
Pairwise Master Key (PMK)	Khóa chủ theo cặp	Root key used to derive session keys
Pairwise Transient Key (PTK)	Khóa tạm thời theo cặp	Session key for unicast traffic
Extensible Authentication Protocol (EAP)	Giao thức xác thực mở rộng	Framework carrying authentication methods
RADIUS	Dịch vụ xác thực người dùng từ xa	AAA protocol commonly used with 802.1X
Virtual Private Network (VPN)	Mạng riêng ảo	Encrypted tunnel over an untrusted network
Rogue AP	AP trái phép	Unauthorised AP on the network
Evil twin	AP song sinh độc hại	Fake AP impersonating a legitimate SSID
Wireless Intrusion Prevention System (WIPS)	Hệ thống ngăn chặn xâm nhập không dây	Detects and mitigates wireless attacks
Deployment and operations		
Site survey	Khảo sát hiện trường	Measuring and modelling coverage on site
Heat map	Bản đồ nhiệt	Map of signal or performance across an area
Coverage hole	Vùng lõm sóng	Area with inadequate signal
Capacity	Dung lượng	Total traffic the network can carry
Cell overlap	Độ chồng lấn ô	Overlap between neighbouring AP coverage areas
Channel plan	Quy hoạch kênh	Assignment of channels to APs
Airtime	Thời gian chiếm kênh	Time a station uses the channel
Channel utilization	Mức sử dụng kênh	Percentage of time the channel is busy
Retry rate	Tỷ lệ truyền lại	Percentage of frames retransmitted
Packet capture	Bắt gói tin	Recording frames for analysis
Monitor mode	Chế độ giám sát	Radio mode capturing frames on a channel
Firmware	Phần mềm điều khiển	Embedded software of a network device

References

[ABV+04] B. Aboba, L. Blunk, J. Vollbrecht, J. Carlson, and H. Levkowetz. Extensible authentication protocol (EAP). RFC 3748, IETF, 2004.

[CMS09a] P. Calhoun, M. Montemurro, and D. Stanley. Control and provisioning of wireless access points (CAPWAP) protocol binding for IEEE 802.11. RFC 5416, IETF, 2009.

[CMS09b] P. Calhoun, M. Montemurro, and D. Stanley. Control and provisioning of wireless access points (CAPWAP) protocol specification. RFC 5415, IETF, 2009.

[FMS01] Scott Fluhrer, Itsik Mantin, and Adi Shamir. Weaknesses in the key scheduling algorithm of RC4. In *Selected Areas in Cryptography (SAC 2001), LNCS 2259*, 2001.

[HK17] D. Harkins and W. Kumari. Opportunistic wireless encryption. RFC 8110, IETF, 2017.

[IEE20] IEEE. IEEE Standard for Local and Metropolitan Area Networks—Port-Based Network Access Control. IEEE Std 802.1X-2020, 2020.

[IEE21] IEEE. IEEE Standard for Information Technology—Telecommunications and Information Exchange between Systems—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications. IEEE Std 802.11-2020 (Revision of IEEE Std 802.11-2016), 2021.

[RWRS00] C. Rigney, S. Willens, A. Rubens, and W. Simpson. Remote authentication dial in user service (RADIUS). RFC 2865, IETF, 2000.

[VP17] Mathy Vanhoef and Frank Piessens. Key reinstallation attacks: Forcing nonce reuse in WPA2. In *Proc. ACM Conference on Computer and Communications Security (CCS)*, 2017.

[VR20] Mathy Vanhoef and Eyal Ronen. Dragonblood: Analyzing the dragonfly handshake of WPA3 and EAP-pwd. In *Proc. IEEE Symposium on Security and Privacy*, 2020.

[Wi-18] Wi-Fi Alliance. WPA3 specification. Wi-Fi Alliance technical specification, 2018.